POWER: USB-C 5V (D101) / LiPo (Q101) auto power path -> VSYS -> U6 LP5907 3.0 V -> +3V0_A (ADS1298 AVDD)
and U9 AP2112K 3.3 V -> +3V3 (ESP32-S3, ADS1298 DVDD). U6 EN = +3V3, so the AFE is off whenever the MCU rail is off.

TEST POINTS

LiPo: J3 = two solder pads for a protected LiPo. U8 MCP73831 charges it from USB at 100 mA (R_PROG 10k).

RLD: 1M || 1.5nF feedback, 330k series to the REF electrode(s) limits DC fault current to <10 uA.

INPUT RC: 22k series (limits fault and plug-insertion current) + 100pF C0G to AGND, ~72 kHz.
Kept small so high-impedance dry electrodes do not turn mains common-mode into differential hum.
ADS1298 oversamples at 512 kHz, so this is the only anti-alias filter needed.

ELECTRODES: J1 TRS for CH1 (T +, R -, S REF).
J6 1 mm FPC carries CH1-CH3 for a flex array or breakout,
in parallel with J1 for CH1: use one or the other.
All REF pins share RLD_ELEC. J6 1/5/9 + MP = AGND guards.

SAFETY: with electrodes on the body and USB-C plugged in, run the laptop on battery,
unplugged from mains, or use a simulated input. Charge while worn only from
an isolated (Class II) charger.
LAYOUT: keep U1 and its supply away from the XIAO antenna.

ADS1298: CH1-CH3 used, CH4-CH8 and TESTP tied to AVDD per datasheet (power down in firmware).
1, 2 or 3 channels by mode, no rework. HPF/LPF in firmware: capture ~0.5 Hz up at 4 kSPS+.
Lead-off detection and RLD channel selection are firmware settings.
Internal clock (CLKSEL high), internal 2.4 V reference.

MCU: ESP32-S3-MINI-1-N8 (PCB antenna, keep copper out of the antenna end). Native USB on IO19/IO20 through USBLC6 ESD to USB-C.
ADS1298 SPI: IO12 SCLK, IO13 DOUT (MISO), IO14 DIN (MOSI), IO15 CS, IO16 DRDY, IO17 START, IO18 RESET.
Flashing over USB uses the built-in USB-Serial-JTAG. If it ever stops enumerating: short TP10 (IO0) to GND, then tap TP9 (EN) to GND.
Charger U8 MCP73831 at 100 mA (R703 10k). 3.3 V rail U9 AP2112K from VSYS.

ON/OFF: SW1 disconnects the battery from the system path (Q101). Off = 0 uA from the cell; USB still powers the board and U8 still charges the cell.
Slider toward pin 1 = ON (connects 2-1). BATTERY MONITOR: VBAT_SYS / 2 into IO10 (ADC1_CH9), 1M/1M + 100nF, 2 uA only while ON.
CHARGE LED: D102 lit while U8 is charging (STAT low); off when full or no USB.

Spec: emg_headband/hardware/emg_afe/README.md
1 to 3 channel surface EMG, ADS1298 + ESP32-S3-MINI-1, JLCPCB assembly

Sheet: /
File: emg_wearable_revA.kicad_sch

Title: EMG wearable AFE

Size: A2	Date: 2026-10-11	Rev: A
KiCad E.D.A. 10.0.7		Id: 1/1